

$$X = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$$

$$X_0 = e^0 = 1$$

$$Y = -\frac{1}{2}\sigma^2 t + \sigma W_t$$

$$\mathbb{E}(Y) = -\frac{1}{2}\sigma^2 t$$

$$V(Y) = \sigma^2 t$$

$$\mathbb{E}(X) = e^{-\frac{1}{2}\sigma^2 t + \frac{1}{2}\sigma^2 t} = e^0 = 1 = X_0$$

$$\rightarrow ds_t/s_t = \underline{\mu dt + \sigma dw} \quad \text{-- IP}$$

$$\rightarrow dB_t/B_t = \underline{r dt}$$

$$d\left(\frac{x}{y}\right) = \left(\frac{x}{y}\right) \left(\frac{dx}{x} - \frac{dy}{y} - \left(\frac{dx}{x}\right)\left(\frac{dy}{y}\right) + \left(\frac{dy}{y}\right)^2 \right)$$

$$d\left(\frac{s}{B}\right) = \left(\frac{s}{B}\right) \left(\frac{ds}{s} - \frac{dB}{B} - \left(\frac{ds}{s}\right)\left(\frac{dB}{B}\right) + \left(\frac{dB}{B}\right)^2 \right)$$

$$= \left(\frac{s}{B}\right) (\mu dt + \sigma dw - r dt)$$

$$= \left(\frac{s}{B}\right) (\underline{\underline{(\mu - r) dt + \sigma dw}}) \quad \text{-- IP}$$

$$\rightarrow \underline{z_t \cdot \sigma d\tilde{w}} \quad \text{-- Q}$$

$$\sigma d\tilde{w} = (\mu - r) dt + \sigma dw$$

$$\Rightarrow d\tilde{w} = \left(\frac{\mu - r}{\sigma}\right) dt + dw$$

$\rightarrow$ (Sharpe Ratio)

$$\frac{dQ}{dP} = e^{-\left(\frac{\mu - r}{\sigma}\right) w_T - \frac{1}{2} \left(\frac{\mu - r}{\sigma}\right)^2 t}$$

at time $(t+dt)$

$$\begin{array}{l} V_t \\ V_{t+dt} \end{array}$$

$$\begin{array}{l} B_t \cdot E_t \\ B_{t+dt} \cdot E_{t+dt} \end{array}$$

$$\begin{aligned} & \frac{\phi_t \cdot S_{t+dt} + \psi_t \cdot B_{t+dt}}{B_{t+dt}} \\ &= B_{t+dt} \left(\phi_t \cdot \frac{S_{t+dt}}{B_{t+dt}} + \psi_t \right) \\ &= B_{t+dt} (\phi_t \cdot Z_{t+dt} + \psi_t) \\ &= B_{t+dt} (\phi_t \cdot (Z_t + dz_t) + \psi_t) \\ &= B_{t+dt} (\underline{E_t} + \phi_t \cdot \underline{dz_t}) = B_{t+dt} \times (\underline{E_t} + \underline{dE_t}) = \underline{B_{t+dt}} \cdot \underline{E_{t+dt}} = \underline{V_{t+dt}} \end{aligned}$$

$$\underline{\underline{\mathbb{E}^Q(\cdot) = \mathbb{E}^P(\cdot) \times \left(\frac{dQ}{dP}\right)}}$$

$$\rightarrow \frac{x_+}{z_+} = \mathbb{E}^N\left(\frac{x_T}{z_T} \mid \mathcal{F}_+\right), \quad \frac{x_+}{u_+} = \mathbb{E}^U\left(\frac{x_T}{u_T} \mid \mathcal{F}_+\right)$$

$$\underline{\underline{z_+ \mathbb{E}^N\left(\frac{x_T}{z_T} \mid \mathcal{F}_+\right) = u_+ \mathbb{E}^U\left(\frac{x_T}{u_T} \mid \mathcal{F}_+\right)}}$$

$$\Rightarrow \underline{\underline{\mathbb{E}^N\left(\frac{x_T}{z_T} \cdot z_+ \mid \mathcal{F}_+\right) = \mathbb{E}^U\left(\frac{x_T}{u_T} \cdot u_+ \mid \mathcal{F}_+\right)}}$$

$$\rightarrow \underline{\underline{\mathbb{E}^{N^U}\left(\frac{x_T}{z_T} \cdot z_+ \times \frac{dQ^N}{dQ^U} \mid \mathcal{F}_+\right) = \mathbb{E}^U\left(\frac{x_T}{u_T} \cdot u_+ \mid \mathcal{F}_+\right)}}$$

$$\left(\frac{z_+}{z_T} \cdot \frac{dQ^N}{dQ^U}\right) = \frac{u_+}{u_T} \Rightarrow \frac{dQ^N}{dQ^U} = \frac{u_+}{u_T} \times \frac{z_T}{z_+} \\ = \frac{z_T}{z_+} \times \frac{u_+}{u_T}$$